



Ahmad Ghalawinji

DATA ENGINEER & AI SPECIALIST

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Data engineer and AI specialist with industry experience building ML pipelines and data systems at scale.
Seeking data engineering or applied AI roles.

Experience

Self-Employed

Remote, France

FREELANCE WEB & INFRASTRUCTURE ENGINEER

Jan. 2025 - Present

- Deployed and managed Linux/WordPress infrastructure for multiple clients with high uptime achieved through proactive monitoring, SSL certificate management, and automated backups.
- Configured Nginx reverse proxies, DNS management, and Git-based deployment workflows to enable rapid iterations and streamlined development processes.
- Technologies:** Linux, Nginx, MySQL, Git, WordPress, SSL/TLS

Amadeus

Nice, France

ARTIFICIAL INTELLIGENCE ENGINEERING INTERN

Feb. 2024 - Aug. 2024

- Improved production forecasting system by designing data pipeline in Databricks to integrate external pricing signals into the machine learning model.
- Achieved 4% improvement in forecast accuracy, resulting in 0.4% revenue increase.
- Technologies:** Python, MongoDB, Azure, Spark on Databricks

Inria

Grenoble, France

MACHINE LEARNING INTERN

Feb. 2023 - Jul. 2023

- Conducted research into deep neural models including transformers and variational autoencoders (VAEs) to enhance representation learning.
- Utilized PyTorch to develop VAE that reduced data dimensionality, enabling study of dynamic system behavior.
- Technologies:** Python, PyTorch, VAEs

Laboratoire d'Informatique de Grenoble (LIG)

Grenoble, France

SOFTWARE ENGINEERING INTERN

Feb. 2022 - Jul. 2022

- Defined automated planning workflows and implemented transition from Domain-Specific Language (DSL) to automated planning using meta-model for Petri nets.
- Technologies:** DSL, Modelling

Education

ENSIMAG and Grenoble Alpes University

Grenoble, France

MSC IN DATA SCIENCE AND ARTIFICIAL INTELLIGENCE (MO SIG M2)

Sep. 2023 - Sep. 2024

- Relevant coursework: Advanced Machine Learning, Natural Language Processing, Big Data Management

ENSIMAG and Grenoble Alpes University

Grenoble, France

MSC IN INFORMATICS (MO SIG M1)

Sep. 2022 - Sep. 2023

- Relevant coursework: Algorithms, Software Engineering, Operating Systems

Tishreen University

Latakia, Syria

BSC IN COMPUTER ENGINEERING

Sep. 2016 - Sep. 2021

- Relevant coursework: Data Structures, Computer Architecture, Database Systems

Skills

Programming

Python (Advanced), SQL, Java

Machine Learning & NLP

Supervised & Unsupervised Learning, NLP, RAG, ChromaDB, VAEs, Transformers, PyTorch, Scikit-learn, XGBoost

Data Engineering

Apache Spark, Apache Airflow, Azure Databricks, AWS (S3, EC2), ETL Pipelines, PostgreSQL, MongoDB, Snowflake, DuckDB, Terraform

Visualization

Streamlit, Plotly, Gradio

Tools & Deployment

Git, Linux, Docker, LaTeX

Projects

TREND ORACLE — NEWS Q&A SYSTEM WITH RAG | GitHub | REPORT

2026

- Built a fully local news Q&A system collecting articles from five international RSS feeds with NLP pipelines for sentiment analysis and named entity recognition.
- Designed retrieval-augmented generation (RAG) architecture using MiniLM embeddings and ChromaDB, enabling natural-language queries over collected news with cited sources.
- Engineered complete data pipeline (collection, processing, indexing) with DuckDB for structured queries and Docker Compose containerization.
- Delivered interactive Gradio interface with streaming responses, clickable source links, and live sidebar for article exploration.
- **Technologies:** Python, spaCy, HuggingFace Transformers, ChromaDB, DuckDB, Ollama, Gradio, Docker

CLIMATE CHANGE & YOUTH CARDIORESPIRATORY FITNESS | GitHub | REPORT

2026

- Investigated climate warming effects on youth cardiorespiratory fitness across 28 European countries using ERA5 climate reanalysis and socioeconomic indicators.
- Trained predictive models (Random Forest, XGBoost) showing 4.7% baseline CRF decline under +3°C warming; mitigation analysis revealed 55% recovery potential through enhanced policies.
- Built interactive Streamlit dashboard for policymakers to explore climate scenarios and evaluate adaptation strategies in real-time.
- **Technologies:** Python, Pandas, NumPy, Scikit-learn, XGBoost, Plotly, Streamlit, ERA5 Climate Data

PORT CALLS DETECTION & VESSEL JOURNEY MODELLING | GitHub | REPORT

2025

- Built a production-style ETL pipeline to detect vessel port call events from 27,000+ AIS records across 16 ports, using geodesic geofencing (20 km threshold) and 30-minute deduplication to minimise false positives.
- Orchestrated a two-task Airflow DAG to extract CSV data into PostgreSQL and detect port entries/exits; inferred next destinations per vessel using a SQL LEAD() window function.
- Improved test coverage with pytest, asserting correct vessel IDs, port names, timestamps, and row values — not just call counts; used tmp_path fixtures for realistic CSV integration tests.
- Identified and documented anchoring false positives; proposed mitigation via speed threshold (<0.5 knots) and positional drift analysis to discriminate anchoring from true port docking.
- Delivered an interactive Streamlit dashboard with multi-dimensional filters and summary metrics; containerised the full stack with Docker Compose.
- **Technologies:** Python, PostgreSQL, Apache Airflow, Streamlit, Docker, Pandas, GeoPy

PRODUCT ANALYTICS PLATFORM — MUSIC STREAMING PIPELINE | GitHub

2025

- Built end-to-end analytics pipeline generating 1M+ synthetic music streaming events across 50,000 users and ingesting them into a partitioned AWS S3 data lake (182 files).
- Loaded raw events into Amazon Redshift with DISTKEY/SORTKEY optimizations; modelled analytical marts (fct_daily_events, fct_daily_active_users, dim_users) for downstream BI consumption.
- Delivered 6-visual Amazon QuickSight dashboard (SPICE engine) covering DAU KPIs, 7-day rolling averages, event distribution by device type, and top-10 event rankings.
- **Technologies:** Python, Faker, AWS S3, Amazon Redshift, Amazon QuickSight

Certificates

- Aug. 2024 **Azure Spark Databricks Essential Training**, LinkedIn Learning
- Aug. 2024 **ETL in Python and SQL**, LinkedIn Learning
- Aug. 2024 **Advanced NoSQL for Data Science**, LinkedIn Learning
- Aug. 2024 **Learning SnowflakeDB**, LinkedIn Learning
- Aug. 2024 **Infrastructure as Code with Terraform**, LinkedIn Learning
- Aug. 2024 **DevOps Foundations: Infrastructure as Code**, LinkedIn Learning
- Aug. 2024 **DevOps Foundations**, LinkedIn Learning
- Aug. 2024 **Agile Foundations**, LinkedIn Learning

Languages

- Arabic** Native
- English** Advanced (C2)
- French** Intermediate (B2)